Intermediate thesis presentation

Analysing Sectoral Unemployment Dynamics in the United States Using Autoregressive Models for Matrix-Valued Time Series

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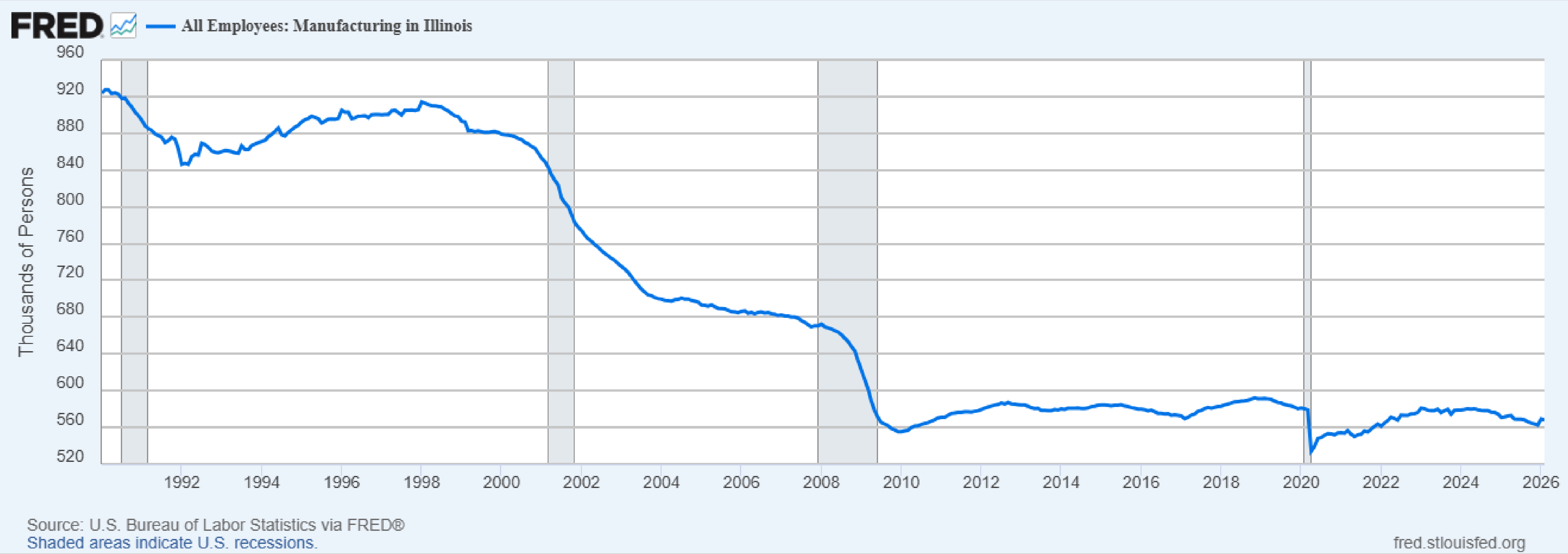
Research question

We want to understand unemployment dynamics across economic sectors and states. To achieve this, we use the Matrix Autoregressive model (MAR) - a parsimonious extension of VAR designed for matrix-valued time series.

First case

1 State 1 Sector (Manufacturing in Illinois)

$$y_t = \beta_0 + \beta_1 y_{t-1} + \epsilon_t$$



All Employees: Manufacturing in Illinois

Second case

1 State 2 Sectors (Manufacturing and Construction in Illinois)

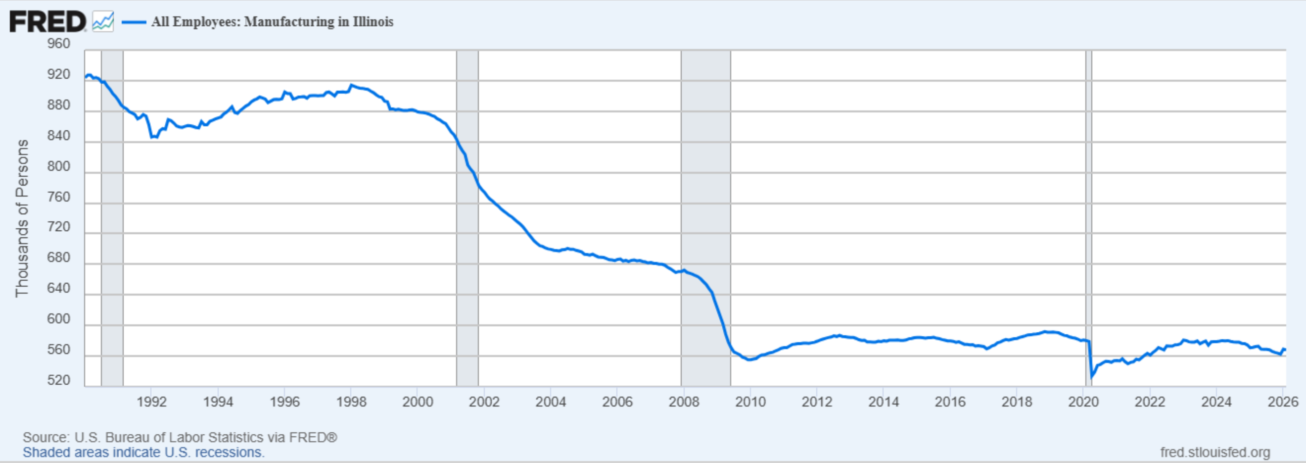
Vector Auto-Regression (VAR) Model - Bivariate VAR(1) — Vector and Matrix Form

Example Bivariate VAR(1) model - (y_t, x_t) and one lag:

$$y_t = a_1 + b_{11} y_{t-1} + b_{12} x_{t-1} + u_t$$
$$x_t = a_2 + b_{21} y_{t-1} + b_{22} x_{t-1} + v_t$$

Matrix Representation

$$\begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} u_t \\ v_t \end{bmatrix}$$



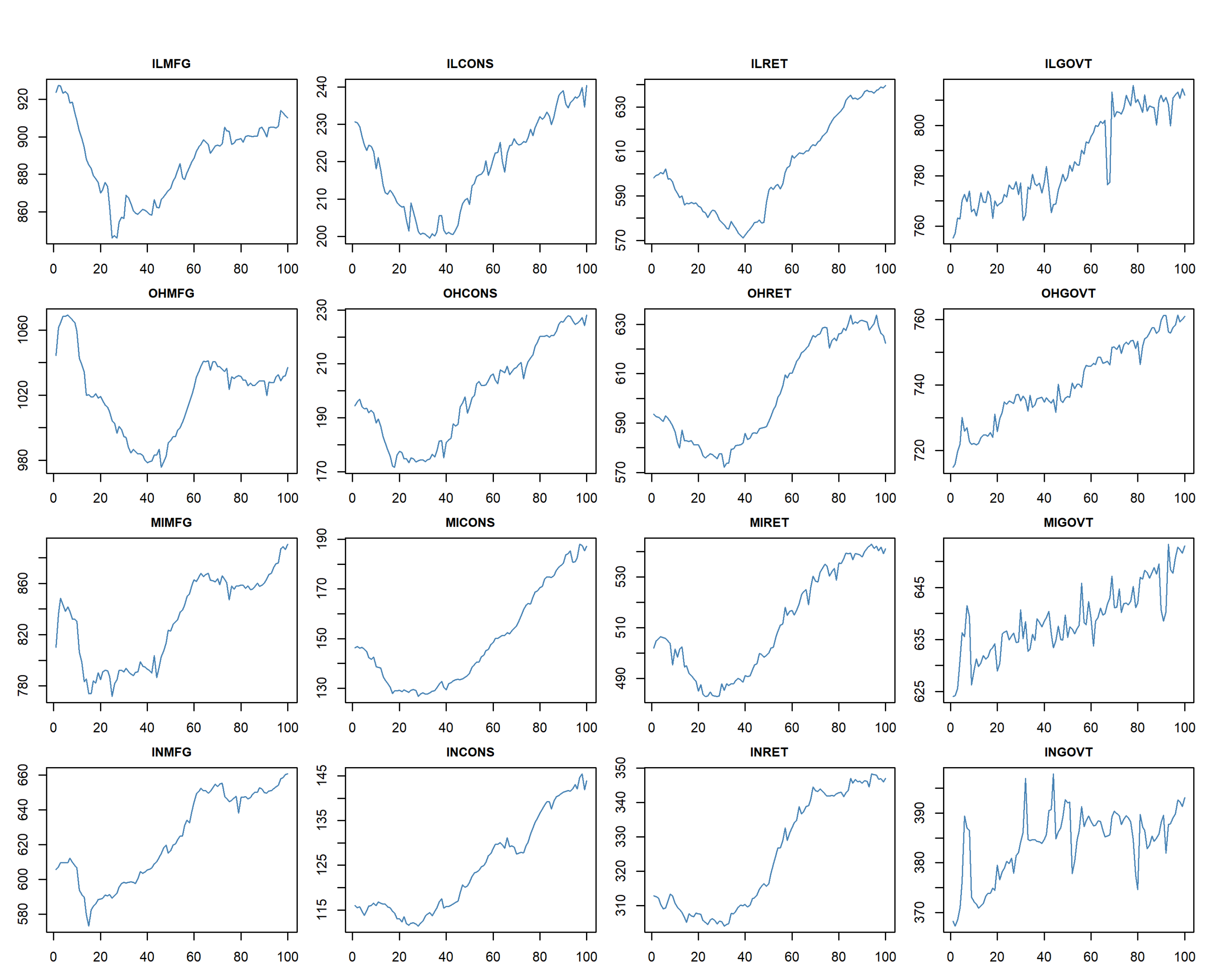
All Employees: Manufacturing in Illinois

All Employees: Construction in Illinois

Third case

- **4 Sectors:**
  - Manufacturing
  - Construction
  - Retail Trade
  - Government
- **4 States:**
  - Illinois
  - Ohio
  - Michigan
  - Indiana
- **Time dimension**

Together, sectors × states × time form a 3-dimensional tensor



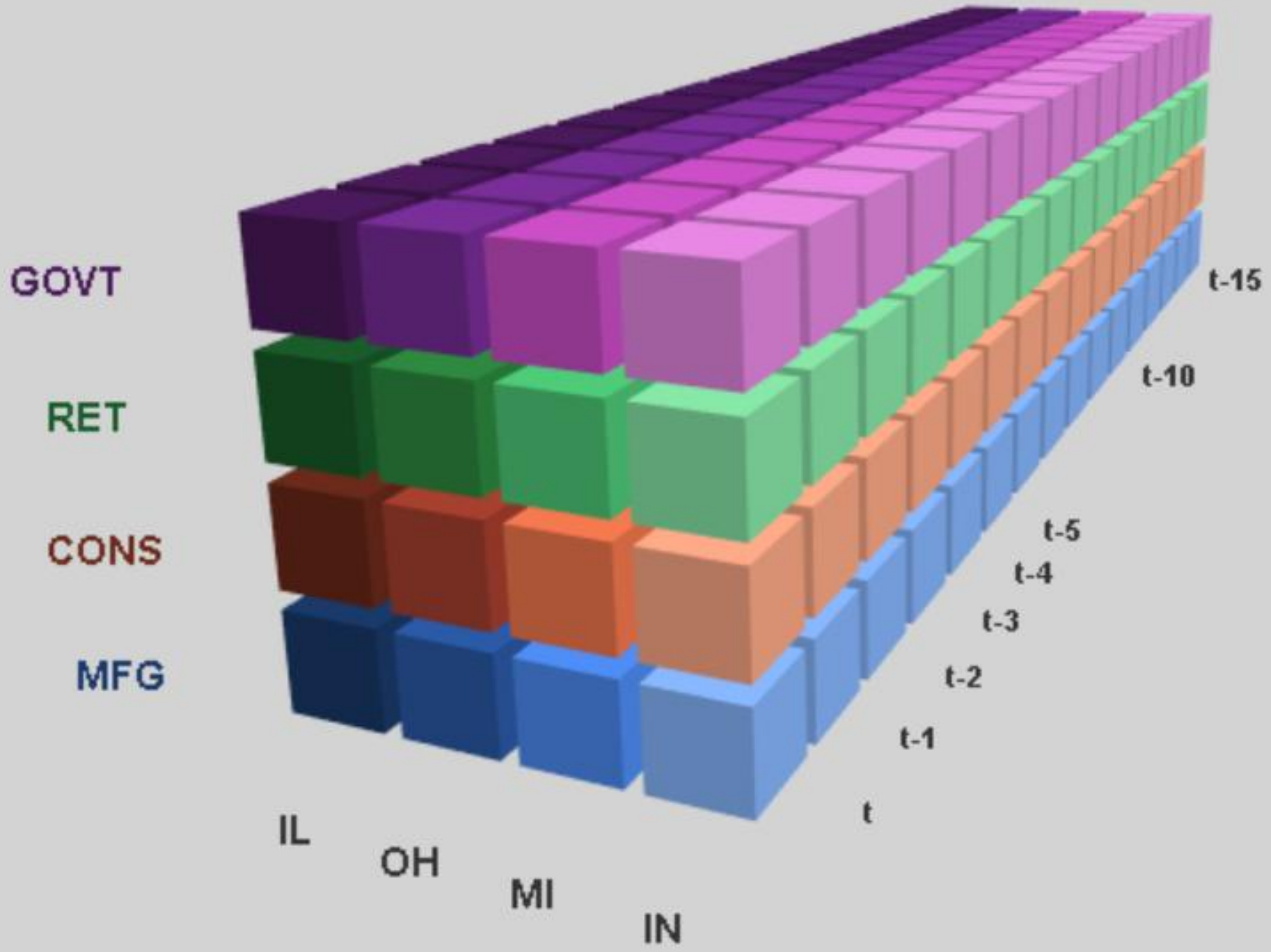
Tensor Format Data

- **Colors represent different sectors:**

  - Purple GOVT – Government
  - Green RET – Retail Trade
  - Orange CONS – Construction
  - Blue MFG – Manufacturing

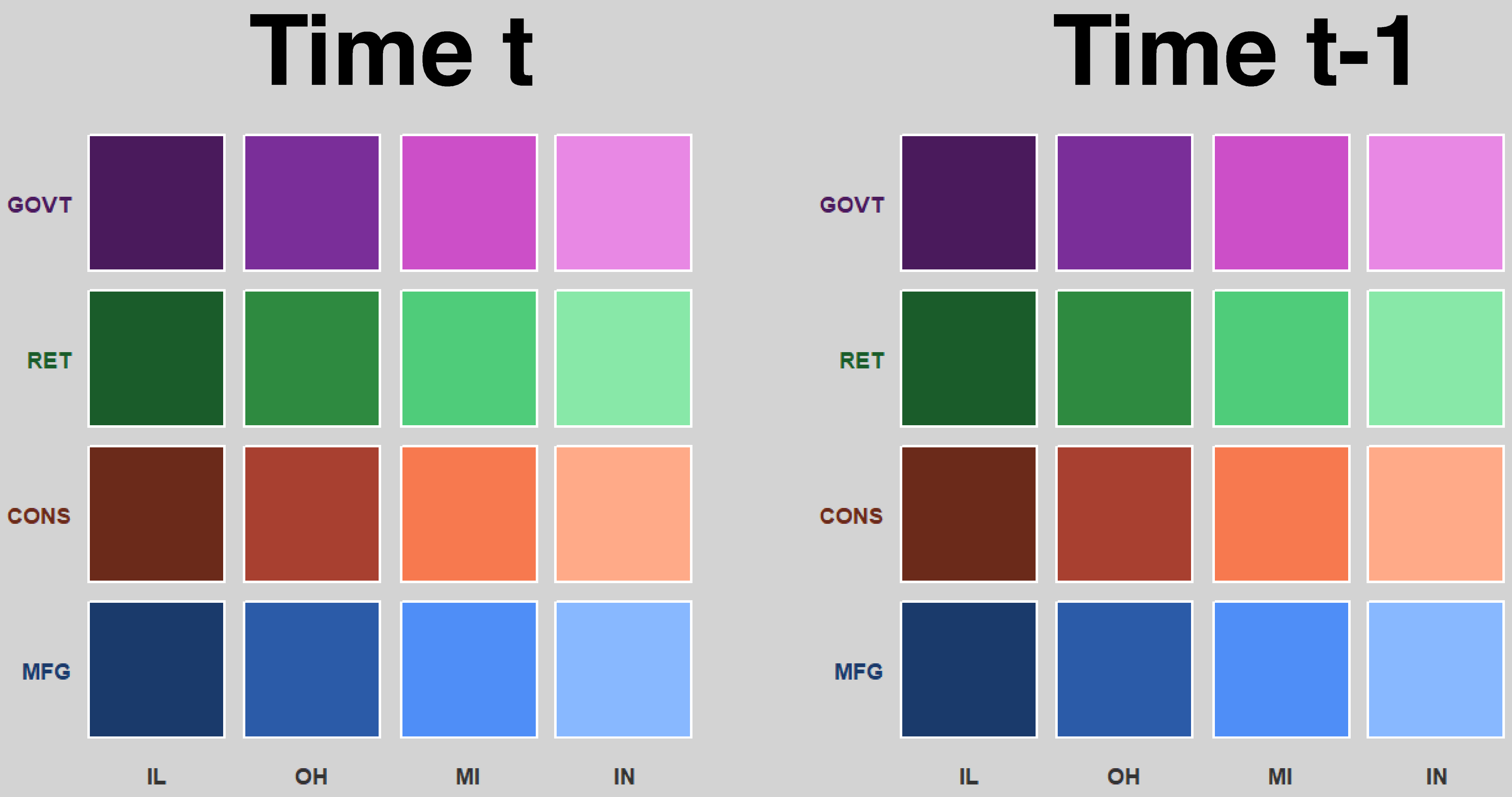
- **Gradient represents different states:**

  - IL – Illinois
  - OH – Ohio
  - MI – Michigan
  - IN – Indiana



What do we want?

- We want to estimate how the employment matrix at time t-1 linearly predicts the matrix at time t, capturing both cross-state and cross-sector dynamics simultaneously.



Vector Auto Regression (VAR)

Example Bivariate VAR(1) model - (y_t, x_t) and one lag:

$$y_t = a_1 + b_{11} y_{t-1} + b_{12} x_{t-1} + u_t$$
$$x_t = a_2 + b_{21} y_{t-1} + b_{22} x_{t-1} + v_t$$

Matrix Representation

$$\begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} u_t \\ v_t \end{bmatrix}$$

Stacking

- Instead of sector/state labels, assign numeric indices to each cell.
- Stack all columns on top of each other to obtain a 16-dimensional vector. This is the standard VAR representation.

Problem?

- The coefficient matrix **B** contains **256 parameters** (16 × 16)
- Our goal is to understand dynamics - not to identify every pairwise state-sector relationship
- This is the **parsimony problem**: too many parameters relative to the signal we seek
- Solution: impose structure by decomposing **B** into two smaller matrices **A** (4 × 4) and **B** (4 × 4), reducing parameters to **32**

$$\begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \\ y_{4t} \\ y_{5t} \\ y_{6t} \\ y_{7t} \\ y_{8t} \\ y_{9t} \\ y_{10t} \\ y_{11t} \\ y_{12t} \\ y_{13t} \\ y_{14t} \\ y_{15t} \\ y_{16t} \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \\ \alpha_6 \\ \alpha_7 \\ \alpha_8 \\ \alpha_9 \\ \alpha_{10} \\ \alpha_{11} \\ \alpha_{12} \\ \alpha_{13} \\ \alpha_{14} \\ \alpha_{15} \\ \alpha_{16} \end{bmatrix} + \begin{bmatrix}
b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} & b_{1,5} & b_{1,6} & b_{1,7} & b_{1,8} & b_{1,9} & b_{1,10} & b_{1,11} & b_{1,12} & b_{1,13} & b_{1,14} & b_{1,15} & b_{1,16} \\
b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} & b_{2,5} & b_{2,6} & b_{2,7} & b_{2,8} & b_{2,9} & b_{2,10} & b_{2,11} & b_{2,12} & b_{2,13} & b_{2,14} & b_{2,15} & b_{2,16} \\
b_{3,1} & b_{3,2} & b_{3,3} & b_{3,4} & b_{3,5} & b_{3,6} & b_{3,7} & b_{3,8} & b_{3,9} & b_{3,10} & b_{3,11} & b_{3,12} & b_{3,13} & b_{3,14} & b_{3,15} & b_{3,16} \\
b_{4,1} & b_{4,2} & b_{4,3} & b_{4,4} & b_{4,5} & b_{4,6} & b_{4,7} & b_{4,8} & b_{4,9} & b_{4,10} & b_{4,11} & b_{4,12} & b_{4,13} & b_{4,14} & b_{4,15} & b_{4,16} \\
b_{5,1} & b_{5,2} & b_{5,3} & b_{5,4} & b_{5,5} & b_{5,6} & b_{5,7} & b_{5,8} & b_{5,9} & b_{5,10} & b_{5,11} & b_{5,12} & b_{5,13} & b_{5,14} & b_{5,15} & b_{5,16} \\
b_{6,1} & b_{6,2} & b_{6,3} & b_{6,4} & b_{6,5} & b_{6,6} & b_{6,7} & b_{6,8} & b_{6,9} & b_{6,10} & b_{6,11} & b_{6,12} & b_{6,13} & b_{6,14} & b_{6,15} & b_{6,16} \\
b_{7,1} & b_{7,2} & b_{7,3} & b_{7,4} & b_{7,5} & b_{7,6} & b_{7,7} & b_{7,8} & b_{7,9} & b_{7,10} & b_{7,11} & b_{7,12} & b_{7,13} & b_{7,14} & b_{7,15} & b_{7,16} \\
b_{8,1} & b_{8,2} & b_{8,3} & b_{8,4} & b_{8,5} & b_{8,6} & b_{8,7} & b_{8,8} & b_{8,9} & b_{8,10} & b_{8,11} & b_{8,12} & b_{8,13} & b_{8,14} & b_{8,15} & b_{8,16} \\
b_{9,1} & b_{9,2} & b_{9,3} & b_{9,4} & b_{9,5} & b_{9,6} & b_{9,7} & b_{9,8} & b_{9,9} & b_{9,10} & b_{9,11} & b_{9,12} & b_{9,13} & b_{9,14} & b_{9,15} & b_{9,16} \\
b_{10,1} & b_{10,2} & b_{10,3} & b_{10,4} & b_{10,5} & b_{10,6} & b_{10,7} & b_{10,8} & b_{10,9} & b_{10,10} & b_{10,11} & b_{10,12} & b_{10,13} & b_{10,14} & b_{10,15} & b_{10,16} \\
b_{11,1} & b_{11,2} & b_{11,3} & b_{11,4} & b_{11,5} & b_{11,6} & b_{11,7} & b_{11,8} & b_{11,9} & b_{11,10} & b_{11,11} & b_{11,12} & b_{11,13} & b_{11,14} & b_{11,15} & b_{11,16} \\
b_{12,1} & b_{12,2} & b_{12,3} & b_{12,4} & b_{12,5} & b_{12,6} & b_{12,7} & b_{12,8} & b_{12,9} & b_{12,10} & b_{12,11} & b_{12,12} & b_{12,13} & b_{12,14} & b_{12,15} & b_{12,16} \\
b_{13,1} & b_{13,2} & b_{13,3} & b_{13,4} & b_{13,5} & b_{13,6} & b_{13,7} & b_{13,8} & b_{13,9} & b_{13,10} & b_{13,11} & b_{13,12} & b_{13,13} & b_{13,14} & b_{13,15} & b_{13,16} \\
b_{14,1} & b_{14,2} & b_{14,3} & b_{14,4} & b_{14,5} & b_{14,6} & b_{14,7} & b_{14,8} & b_{14,9} & b_{14,10} & b_{14,11} & b_{14,12} & b_{14,13} & b_{14,14} & b_{14,15} & b_{14,16} \\
b_{15,1} & b_{15,2} & b_{15,3} & b_{15,4} & b_{15,5} & b_{15,6} & b_{15,7} & b_{15,8} & b_{15,9} & b_{15,10} & b_{15,11} & b_{15,12} & b_{15,13} & b_{15,14} & b_{15,15} & b_{15,16} \\
b_{16,1} & b_{16,2} & b_{16,3} & b_{16,4} & b_{16,5} & b_{16,6} & b_{16,7} & b_{16,8} & b_{16,9} & b_{16,10} & b_{16,11} & b_{16,12} & b_{16,13} & b_{16,14} & b_{16,15} & b_{16,16}
\end{bmatrix} \times \begin{bmatrix} y_{1t-1} \\ y_{2t-1} \\ y_{3t-1} \\ y_{4t-1} \\ y_{5t-1} \\ y_{6t-1} \\ y_{7t-1} \\ y_{8t-1} \\ y_{9t-1} \\ y_{10t-1} \\ y_{11t-1} \\ y_{12t-1} \\ y_{13t-1} \\ y_{14t-1} \\ y_{15t-1} \\ y_{16t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \\ \varepsilon_{3t} \\ \varepsilon_{4t} \\ \varepsilon_{5t} \\ \varepsilon_{6t} \\ \varepsilon_{7t} \\ \varepsilon_{8t} \\ \varepsilon_{9t} \\ \varepsilon_{10t} \\ \varepsilon_{11t} \\ \varepsilon_{12t} \\ \varepsilon_{13t} \\ \varepsilon_{14t} \\ \varepsilon_{15t} \\ \varepsilon_{16t} \end{bmatrix}$$

$$\begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} u_t \\ v_t \end{bmatrix}$$

From VAR to MAR: Structured Coefficient Decomposition

Example Bivariate VAR(1) model - (y_t, x_t) and one lag:

$$y_t = a_1 + b_{11} y_{t-1} + b_{12} x_{t-1} + u_t$$
$$x_t = a_2 + b_{21} y_{t-1} + b_{22} x_{t-1} + v_t$$

$$\boldsymbol{Y_t = AY_{t-1}B' + E_t}$$

Matrix Representation

$$\begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} u_t \\ v_t \end{bmatrix}$$

Coefficient matrix B from VAR has the same purpose as matrices A and B from MAR
Matrix A explain the column (state) interaction while matrix B explains the row (sector) dynamics

MAR

$$Y_t = AY_{t-1}B' + E_t$$

$$\underbrace{\begin{bmatrix} y_{1t} & y_{5t} & y_{9t} & y_{13t} \\ y_{2t} & y_{6t} & y_{10t} & y_{14t} \\ y_{3t} & y_{7t} & y_{11t} & y_{15t} \\ y_{4t} & y_{8t} & y_{12t} & y_{16t} \end{bmatrix}}_{Y_t} = \underbrace{\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{bmatrix}}_{A} \times \underbrace{\begin{bmatrix} y_{1t-1} & y_{5t-1} & y_{9t-1} & y_{13t-1} \\ y_{2t-1} & y_{6t-1} & y_{10t-1} & y_{14t-1} \\ y_{3t-1} & y_{7t-1} & y_{11t-1} & y_{15t-1} \\ y_{4t-1} & y_{8t-1} & y_{12t-1} & y_{16t-1} \end{bmatrix}}_{Y_{t-1}} \times \underbrace{\begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} \\ b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} \\ b_{3,1} & b_{3,2} & b_{3,3} & b_{3,4} \\ b_{4,1} & b_{4,2} & b_{4,3} & b_{4,4} \end{bmatrix}}_{B'} + \underbrace{\begin{bmatrix} \varepsilon_{1,1,t} & \varepsilon_{1,2,t} & \varepsilon_{1,3,t} & \varepsilon_{1,4,t} \\ \varepsilon_{2,1,t} & \varepsilon_{2,2,t} & \varepsilon_{2,3,t} & \varepsilon_{2,4,t} \\ \varepsilon_{3,1,t} & \varepsilon_{3,2,t} & \varepsilon_{3,3,t} & \varepsilon_{3,4,t} \\ \varepsilon_{4,1,t} & \varepsilon_{4,2,t} & \varepsilon_{4,3,t} & \varepsilon_{4,4,t} \end{bmatrix}}_{E_t}$$

Interpreting Matrices A and B

- **A** answers: *"Did high employment in state X last month predict high employment in state Y this month?"* - it captures **cross-state spillovers**

- **B** answers: *"Did high employment in sector X last month predict high employment in sector Y this month?"* - it captures **cross-sector co-movement**

- A large **off-diagonal entry** in A or B means two states (or sectors) strongly influence each other

- A near-**diagonal** matrix means each state/sector mostly evolves on its own

$$\begin{matrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{matrix}$$

A

$$\begin{matrix} b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} \\ b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} \\ b_{3,1} & b_{3,2} & b_{3,3} & b_{3,4} \\ b_{4,1} & b_{4,2} & b_{4,3} & b_{4,4} \end{matrix}$$

B'

Estimation of A and B: 3 main methods

- **Projection method (PROJ):** Estimate the full VAR first, then project it onto the Kronecker-product form. It is **easy to understand** and has a closed-form SVD step, but it is usually **less accurate**.

- **Iterated least squares (LSE):** Estimate A and B directly by minimizing the MAR fitting error. It is more **natural for the model** and usually performs **better than PROJ**, but it requires an **iterative algorithm**.

- **MLE with structured covariance (MLEs):** Jointly estimate A, B, and the row/column covariance structure. It is the most **statistically efficient** when that covariance assumption is correct, but also the most **complex** method.

Estimation of A and B using LSE

Overall Goal: Minimize E_t

$$
\underbrace{\begin{bmatrix} y_{1t} & y_{5t} & y_{9t} & y_{13t} \\ y_{2t} & y_{6t} & y_{10t} & y_{14t} \\ y_{3t} & y_{7t} & y_{11t} & y_{15t} \\ y_{4t} & y_{8t} & y_{12t} & y_{16t} \end{bmatrix}}_{Y_t}
=
\underbrace{\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{bmatrix}}_{A}
\times
\underbrace{\begin{bmatrix} y_{1t-1} & y_{5t-1} & y_{9t-1} & y_{13t-1} \\ y_{2t-1} & y_{6t-1} & y_{10t-1} & y_{14t-1} \\ y_{3t-1} & y_{7t-1} & y_{11t-1} & y_{15t-1} \\ y_{4t-1} & y_{8t-1} & y_{12t-1} & y_{16t-1} \end{bmatrix}}_{Y_{t-1}}
\times
\underbrace{\begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} \\ b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} \\ b_{3,1} & b_{3,2} & b_{3,3} & b_{3,4} \\ b_{4,1} & b_{4,2} & b_{4,3} & b_{4,4} \end{bmatrix}}_{B'}
+
\underbrace{\begin{bmatrix} \varepsilon_{1,1,t} & \varepsilon_{1,2,t} & \varepsilon_{1,3,t} & \varepsilon_{1,4,t} \\ \varepsilon_{2,1,t} & \varepsilon_{2,2,t} & \varepsilon_{2,3,t} & \varepsilon_{2,4,t} \\ \varepsilon_{3,1,t} & \varepsilon_{3,2,t} & \varepsilon_{3,3,t} & \varepsilon_{3,4,t} \\ \varepsilon_{4,1,t} & \varepsilon_{4,2,t} & \varepsilon_{4,3,t} & \varepsilon_{4,4,t} \end{bmatrix}}_{E_t}
$$

Estimation of A and B using LSE

Overall Goal: Minimize E_t

Step 1. Choose some specific matrix A and fix it

Step 2. Calculate matrix B

$$\underset{Y_t}{\begin{bmatrix} y_{1t} & y_{5t} & y_{9t} & y_{13t} \\ y_{2t} & y_{6t} & y_{10t} & y_{14t} \\ y_{3t} & y_{7t} & y_{11t} & y_{15t} \\ y_{4t} & y_{8t} & y_{12t} & y_{16t} \end{bmatrix}} = \underset{A}{\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{bmatrix}} \times \underset{Y_{t-1}}{\begin{bmatrix} y_{1t-1} & y_{5t-1} & y_{9t-1} & y_{13t-1} \\ y_{2t-1} & y_{6t-1} & y_{10t-1} & y_{14t-1} \\ y_{3t-1} & y_{7t-1} & y_{11t-1} & y_{15t-1} \\ y_{4t-1} & y_{8t-1} & y_{12t-1} & y_{16t-1} \end{bmatrix}} \times \underset{B'}{\begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} \\ b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} \\ b_{3,1} & b_{3,2} & b_{3,3} & b_{3,4} \\ b_{4,1} & b_{4,2} & b_{4,3} & b_{4,4} \end{bmatrix}} + \underset{E_t}{\begin{bmatrix} \varepsilon_{1,1,t} & \varepsilon_{1,2,t} & \varepsilon_{1,3,t} & \varepsilon_{1,4,t} \\ \varepsilon_{2,1,t} & \varepsilon_{2,2,t} & \varepsilon_{2,3,t} & \varepsilon_{2,4,t} \\ \varepsilon_{3,1,t} & \varepsilon_{3,2,t} & \varepsilon_{3,3,t} & \varepsilon_{3,4,t} \\ \varepsilon_{4,1,t} & \varepsilon_{4,2,t} & \varepsilon_{4,3,t} & \varepsilon_{4,4,t} \end{bmatrix}}$$

Estimation of A and B using LSE

Overall Goal: Minimize E_t

Step 3. Fix matrix B

Step 4. Recalculate matrix A

$$
\underbrace{\begin{bmatrix} y_{1t} & y_{5t} & y_{9t} & y_{13t} \\ y_{2t} & y_{6t} & y_{10t} & y_{14t} \\ y_{3t} & y_{7t} & y_{11t} & y_{15t} \\ y_{4t} & y_{8t} & y_{12t} & y_{16t} \end{bmatrix}}_{Y_t}
=
\underbrace{\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{bmatrix}}_{A}
\times
\underbrace{\begin{bmatrix} y_{1t-1} & y_{5t-1} & y_{9t-1} & y_{13t-1} \\ y_{2t-1} & y_{6t-1} & y_{10t-1} & y_{14t-1} \\ y_{3t-1} & y_{7t-1} & y_{11t-1} & y_{15t-1} \\ y_{4t-1} & y_{8t-1} & y_{12t-1} & y_{16t-1} \end{bmatrix}}_{Y_{t-1}}
\times
\underbrace{\begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} \\ b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} \\ b_{3,1} & b_{3,2} & b_{3,3} & b_{3,4} \\ b_{4,1} & b_{4,2} & b_{4,3} & b_{4,4} \end{bmatrix}}_{B'}
+
\underbrace{\begin{bmatrix} \varepsilon_{1,1,t} & \varepsilon_{1,2,t} & \varepsilon_{1,3,t} & \varepsilon_{1,4,t} \\ \varepsilon_{2,1,t} & \varepsilon_{2,2,t} & \varepsilon_{2,3,t} & \varepsilon_{2,4,t} \\ \varepsilon_{3,1,t} & \varepsilon_{3,2,t} & \varepsilon_{3,3,t} & \varepsilon_{3,4,t} \\ \varepsilon_{4,1,t} & \varepsilon_{4,2,t} & \varepsilon_{4,3,t} & \varepsilon_{4,4,t} \end{bmatrix}}_{E_t}
$$

Estimation of A and B using LSE

Overall Goal: Minimize E_t

Step 5: Repeat iterating until the change in the objective function falls below a convergence threshold ε

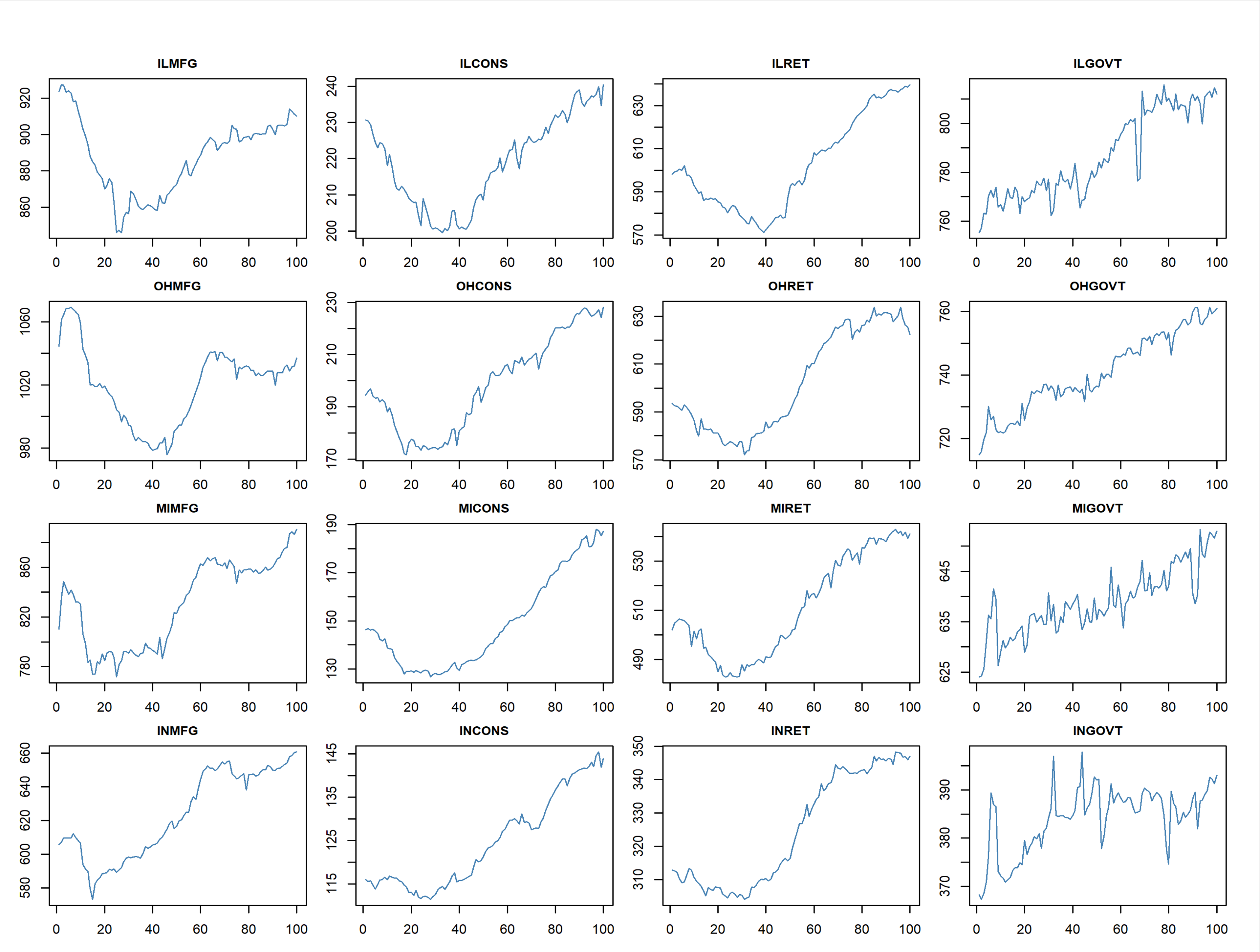
$$
\underbrace{\begin{bmatrix} y_{1t} & y_{5t} & y_{9t} & y_{13t} \\ y_{2t} & y_{6t} & y_{10t} & y_{14t} \\ y_{3t} & y_{7t} & y_{11t} & y_{15t} \\ y_{4t} & y_{8t} & y_{12t} & y_{16t} \end{bmatrix}}_{Y_t}
=
\underbrace{\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{bmatrix}}_{A}
\times
\underbrace{\begin{bmatrix} y_{1t-1} & y_{5t-1} & y_{9t-1} & y_{13t-1} \\ y_{2t-1} & y_{6t-1} & y_{10t-1} & y_{14t-1} \\ y_{3t-1} & y_{7t-1} & y_{11t-1} & y_{15t-1} \\ y_{4t-1} & y_{8t-1} & y_{12t-1} & y_{16t-1} \end{bmatrix}}_{Y_{t-1}}
\times
\underbrace{\begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} \\ b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} \\ b_{3,1} & b_{3,2} & b_{3,3} & b_{3,4} \\ b_{4,1} & b_{4,2} & b_{4,3} & b_{4,4} \end{bmatrix}}_{B'}
+
\underbrace{\begin{bmatrix} \varepsilon_{1,1,t} & \varepsilon_{1,2,t} & \varepsilon_{1,3,t} & \varepsilon_{1,4,t} \\ \varepsilon_{2,1,t} & \varepsilon_{2,2,t} & \varepsilon_{2,3,t} & \varepsilon_{2,4,t} \\ \varepsilon_{3,1,t} & \varepsilon_{3,2,t} & \varepsilon_{3,3,t} & \varepsilon_{3,4,t} \\ \varepsilon_{4,1,t} & \varepsilon_{4,2,t} & \varepsilon_{4,3,t} & \varepsilon_{4,4,t} \end{bmatrix}}_{E_t}
$$

What should we do?

- We estimate A and B using the **TensorTS** package in R
- Before fitting the model, data must be preprocessed into **3D tensor format** (sectors × states × time)
- Preprocessing steps: **First Differencing** (to achieve stationarity) and **ADF stability check**
- Note: the BLS data reports **employed persons**, not unemployment counts - first differences approximate month-over-month employment changes

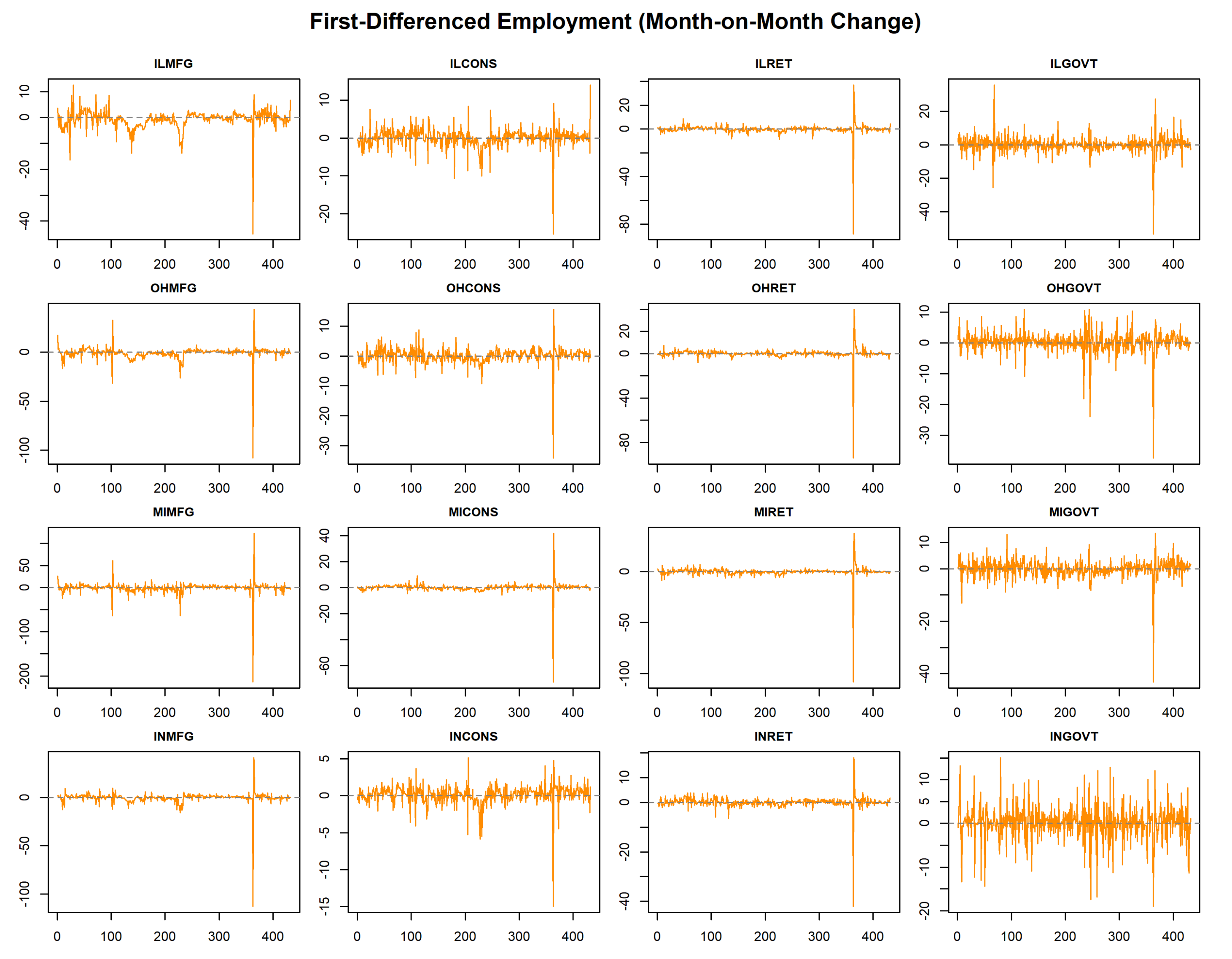
Preprocessing our data

- This is how our initial data looks like
- Just by looking at it we can already say that it is non-stationary
- Non-stationary data violates the stability conditions required by the MAR model
- We apply **First Differencing**: $\Delta y_t = y_t - y_{t-1}$ to remove trends and approximate stationarity

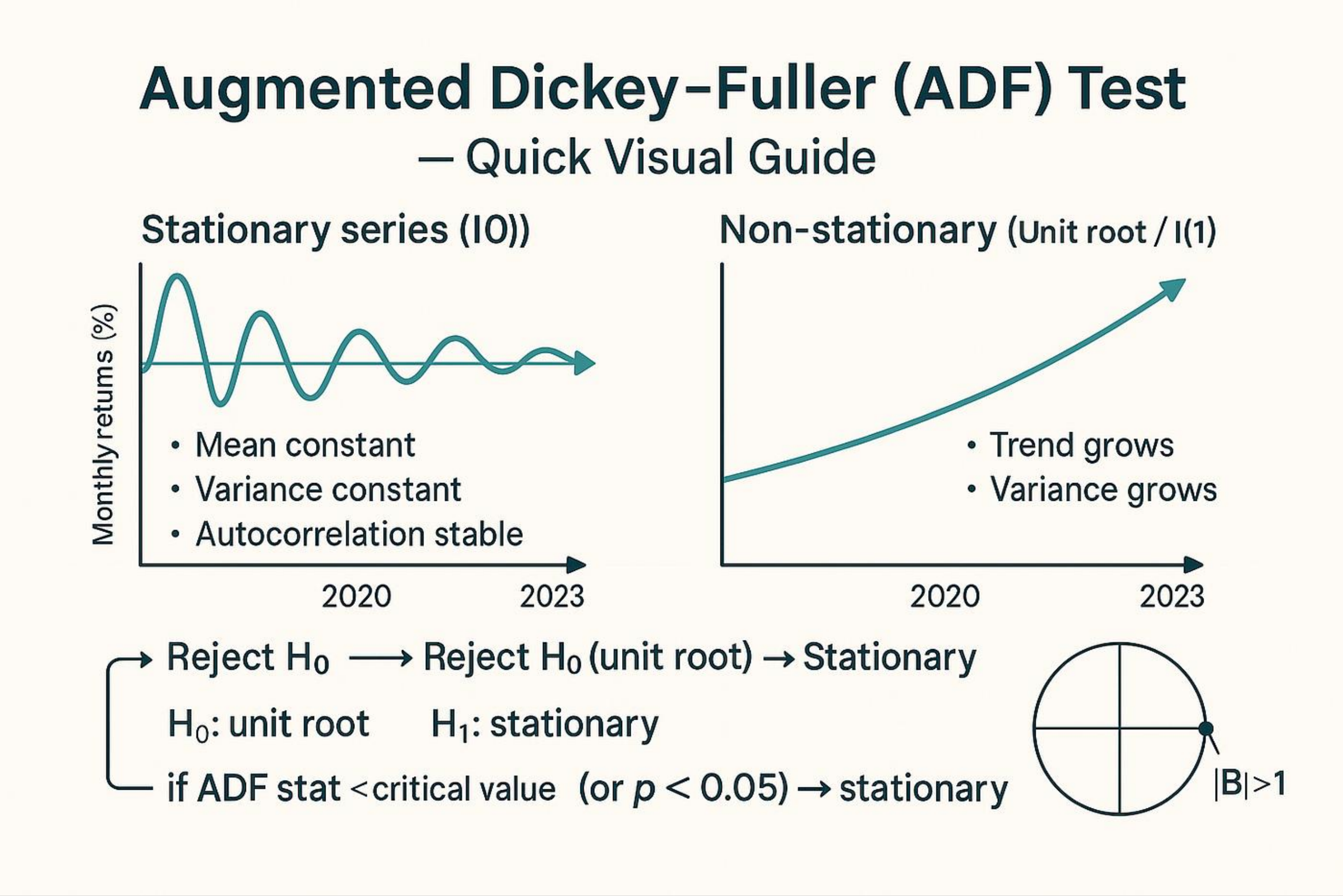


Preprocessing our data

- After first differencing, series fluctuate around zero with no visible trend - consistent with stationarity
- Formal confirmation requires the **Augmented Dickey-Fuller (ADF) test**
- ADF null hypothesis: the series has a **unit root** (non-stationary); rejection confirms stationarity



Preprocessing our data



Stability check using ADF

```
> # ══════════════════════════════════════════
> # STEP 3: ADF TESTS ON DIFFERENCED DATA
> # ═══════════════════════ .... [TRUNCATED]

=== ADF Tests: Differenced Data ===

> for (i in seq_along(states)) {
+   for (j in seq_along(sectors)) {
+     result <- adf.test(xx_diff[, i, j])
+     cat(states[i], sectors[j], "| p = ..."
IL MFG | p = 0.01 -> stationary
IL CONS | p = 0.01 -> stationary
IL RET | p = 0.01 -> stationary
IL GOVT | p = 0.01 -> stationary
IL FIRE | p = 0.01 -> stationary
OH MFG | p = 0.01 -> stationary
OH CONS | p = 0.01 -> stationary
OH RET | p = 0.01 -> stationary
OH GOVT | p = 0.01 -> stationary
OH FIRE | p = 0.01 -> stationary
MI MFG | p = 0.01 -> stationary
MI CONS | p = 0.01 -> stationary
MI RET | p = 0.01 -> stationary
MI GOVT | p = 0.01 -> stationary
MI FIRE | p = 0.01 -> stationary
IN MFG | p = 0.01 -> stationary
IN CONS | p = 0.01 -> stationary
IN RET | p = 0.01 -> stationary
IN GOVT | p = 0.01 -> stationary
IN FIRE | p = 0.01 -> stationary
KY MFG | p = 0.01 -> stationary
KY CONS | p = 0.01 -> stationary
KY RET | p = 0.01 -> stationary
KY GOVT | p = 0.01 -> stationary
KY FIRE | p = 0.01 -> stationary
```

R output ADF

Results for A and B using LSE

Matrix A — State Dynamics

	IL	OH	MI	IN	KY
IL	0.111	0.078	0.056	-0.049	0.073
OH	0.124	0.150	0.258	-0.095	-0.031
MI	0.278	0.215	0.752	-0.206	-0.186
IN	0.182	0.142	0.103	-0.032	0.056
KY	0.041	0.047	0.125	-0.038	-0.002

Coeff. 0.75 0.50 0.25 0.00

Matrix B — Sector Dynamics

	MFG	CONS	RET	GOVT	FIRE
MFG	-0.206	1.875	-0.307	-0.805	-0.712
CONS	-0.009	0.099	-0.170	-0.193	0.070
RET	0.026	0.622	-0.248	-0.393	-0.297
GOVT	0.033	0.013	0.109	-0.054	0.133
FIRE	-0.003	0.031	-0.003	-0.030	0.156

Coeff. 1 0

Future work

- **Eigenvalue check** - verify that the spectral radius of $\mathbf{A} \otimes \mathbf{B}$ is less than 1, confirming the model is stable and stationary
- **Residual check** - inspect residual matrices for autocorrelation to confirm the model captures the dynamics adequately
- **Compare 3 estimation methods** - run PROJ, LSE, and MLEs on our data and compare the estimated **A** and **B** matrices across methods to see how sensitive results are to the choice of estimator
- **Economic interpretation** - interpret the off-diagonal entries of **A** and **B** to identify which states and sectors most strongly influence each other (e.g., does a drop in Illinois manufacturing spill over into Ohio?)
- **Extend to MAR(p)** - test whether adding a second lag MAR(2) improves model fit, capturing slower-moving employment dynamics
- **Out-of-sample forecasting** - use the fitted MAR model to predict next month's employment matrix and compare forecast accuracy against a simple VAR benchmark

Thank you for your attention!

Q&A